

1. Consider the space of polynomials of degree ≤ 3 with inner product $\langle p, q \rangle = \int_{-1}^1 p(x)q(x)dx$. Apply Gram-Schmidt to the monomial basis to obtain an orthogonal basis for the space. The resulting polynomials are called Legendre polynomials.
2. Consider the space V of functions f that are continuous on $[-\pi, \pi]$, with subspace W spanned by

$$\cos(nx), n = 0, \dots, N; \sin(nx), n = 1, \dots, N.$$

Find an expression for the orthogonal projection of a function $f \in V$ onto W , using the L^2 inner product.

3. Let $\mathbf{A}$ be a real skew-symmetric matrix. Without referring to eigenvalues, prove that

$$\mathbf{Q} = (\mathbf{I} - \mathbf{A})(\mathbf{I} + \mathbf{A})^{-1}$$

is

- (a) Well-defined (i.e. the inverse exists), and
- (b) An orthogonal matrix.